Adjustable 1-10 Minute Timer Project

THE ELECTRONICS CLUB

A kit for this project is available from RSH Electronics.

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The circuit starts timing when switched on. The green LED lights to show that timing is in progress. When the time period is over the green LED turns off, the red LED turns on and the bleeper sounds.

The time period is set by adjusting the variable resistor. It can be adjusted from 1 to 10 minutes (approximately) with the parts shown in the diagram. You can mark the times on a scale drawn on the box.

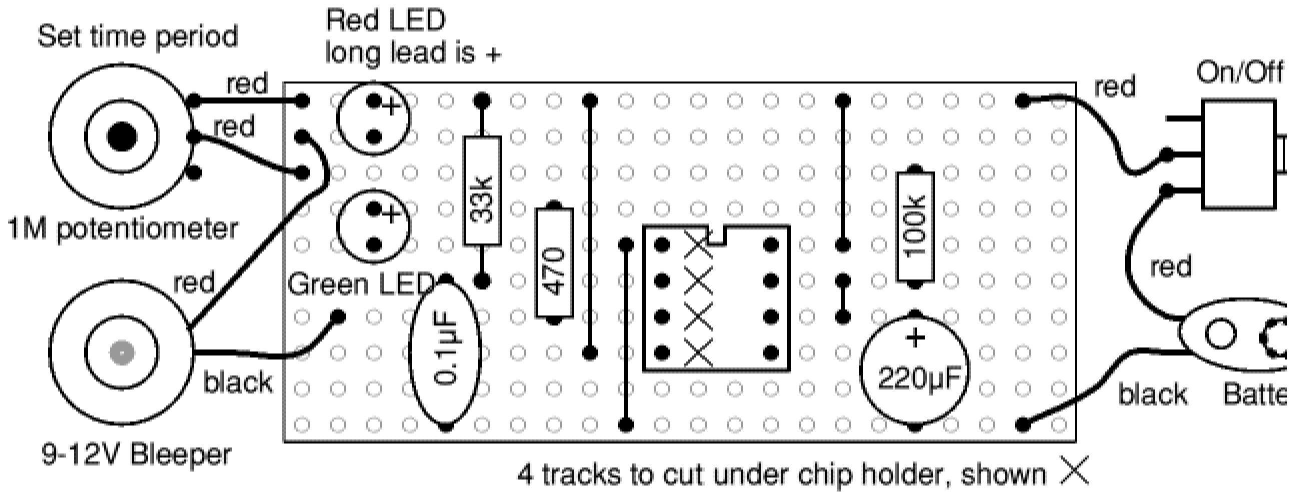
Please note that the range of time periods is only approximate. With perfect components the maximum time period should be 4½ minutes, but this is typically extended to about 10 minutes because the 220µF timing capacitor slowly leaks charge. This is a problem with all electrolytic capacitors, but some leak more than others. In addition the actual value of electrolytic capacitors can vary by as much as ±30% of their rated value.

This project uses a power-on triggered 555 monostable circuit.

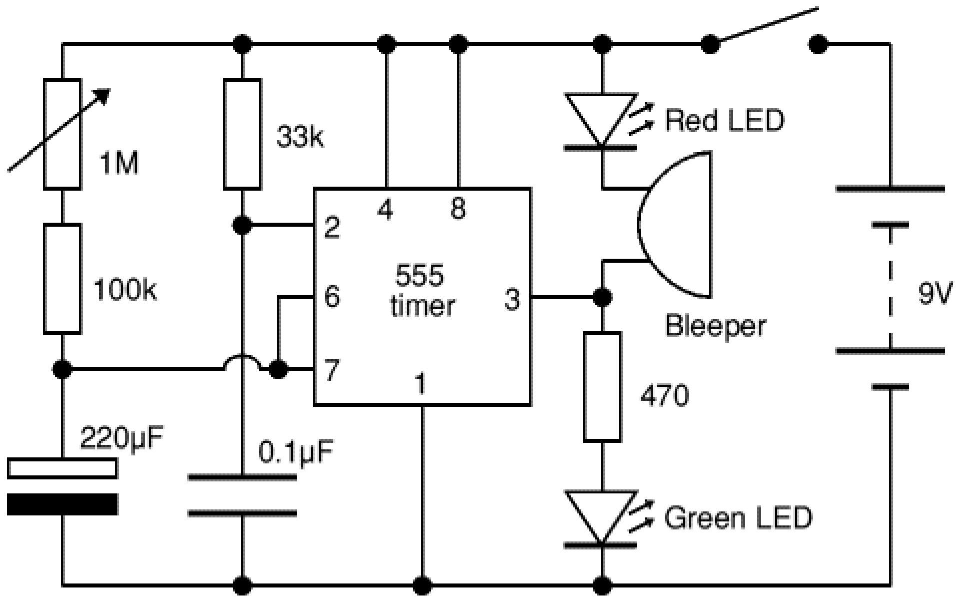
Parts Required

- resistors: 470, 33k, 100k
- variable resistor: 1M
- capacitors: 0.1µF, 220µF 16V radial
- LEDs: red, green
- bleeper 9-12V
- 555 timer IC
- 8-pin DIL socket for IC
- on/off switch
- battery clip for 9V PP3
- stripboard 10 rows × 22 holes

Stripboard Layout



Circuit diagram



A kit for this project is available from RSH Electronics. If you are new to electronics buying a kit is a good way to be sure you have the correct parts for the project.

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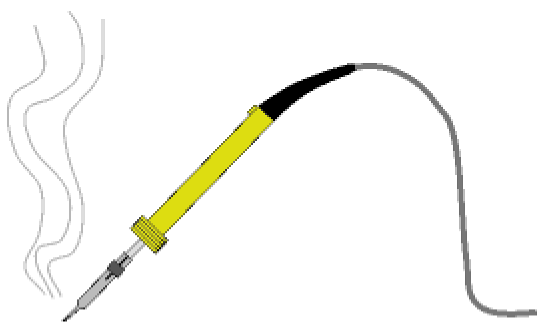

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